CS-2008 Numerical Computing **Quiz 1 - BCS-7A** **Fall 2025**

Name:_____ Roll Number: _____

Q1. Write down the difference between single precision and double precision

(2 + 4 + 2)

a) Explain the IEEE-754 floating-point representation for single precision (32-bit) and double precision (64-bit).

b) How many significant digits (decimal precision) can each representation (single and double precision) accurately store?

Q2. Explain how NaN (Not a Number) is represented in the IEEE-754 floating-point standard. **2**

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Q3. List and briefly explain the different types of errors that occur in numerical computation (with example) **5**

Q4. If x = 0.3721448693 and x*=0.37202 what is the relative error in the computation of x with four decimal digits of accuracy? **2**

Q5: Using the Bisection Method, perform two iterations to approximate the root of the following function within the interval [a,b]: **3**

f(x) = x^3 + x - 3 with the initial interval [a,b]=[0,4]

For each iteration, show the values of a, b, c=a+b/2, and f(c).

Finally, after two iterations, specify the new interval where the root of the function lies.

Iter#	a	b	c	f(a)	f(b)	f(c)